IE 549 — Machine Vision In Intelligent Robotic Systems

HOMEWORK #2: OJECT RECOGNITION

Due on Tuesday, March 12, 2026 (by Gradescope)

Theoretic Question Implementation Question Bonus Question

1. (50%) During the surgery, a human scrub nurse is responsible for retrieve the surgical instruments that remain in the surgical zone for extended periods of time. Those instruments are usually replaced by new ones, waiting to be used in the Mayo tray. The only assumption that can be made is that the instruments are not overlapping. This scenario is realistic since the surgeon usually keeps a very small number of instruments (2-3) around the patient to avoid leaving those instruments behind (retained instruments). That is, almost every time that the surgeon is done with the use of an instrument, the nurse picks it from the surgical site.

Your objective is test both the BRIEF and ORB object descriptors (which are very similar to SIFT) to describe uniquely the specific surgical instrument. Given one of the images at random from the dataset, the algorithm must recognize each instrument, draw a bounding box around it, the major and minor axis, the centroid and the name of the instrument (see figure). You have to also follow the k-cross fold method to train/test to assess your algorithm, and select the method that works the best (BRIEF or ORB). <u>The testing images will not be given to you</u>.

You need to upload the code (OpenCV), with comments explaining every step. You need to create a Python function, called "find_instruments" which receives as a parameter a filename. This filename stands for an image file located in the same folder where the script file is residing (must be a color image). We will provide with the training dataset of 20 images. Also you need to include functions to do the following:

1) (25%) Train/test your method using k-cross folding. Assume k = total images.
2) (25%) Code to print the confusion matrix

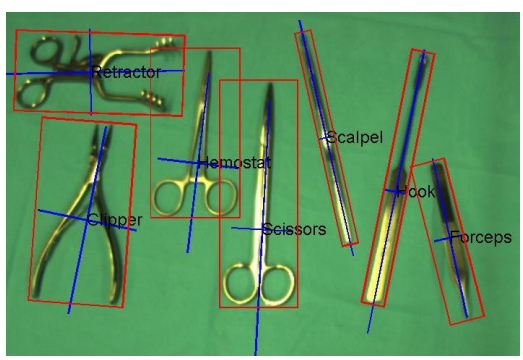

Figure 1. Example of the output of the function "find_instruments. with the argument 1.jpg".

2. (50%) In this exercise, we are interested in building a background from a video. The goal is to create a background model from a sequence of frames. Unfortunately, in every frame we have some elements that are “foreground” which affects the ability of using a traditional technique for background modeling. However, we are lucky because this “foreground” is always moving, and therefore we can rely that for every pixel in the background we will have the chance to observe it at some point in time. You cannot use the frames in the first 26 seconds to build the background alone, neither the last 42 seconds. You can only use the video segment when there is some foreground in the video.

The scene is a view inside a laboratory by one camera view.: https://documents.epfl.ch/groups/c/cv/cvlab-pom-video1/www/6p-c0.avi

You need to write a script that “recover” the background without any students. The script takes a video as an input (the video in the top link) and outputs an image of the background without the students.

You should use different techniques related to background modeling and some simple rules. A solution that takes the overall average is not considered sufficient.

You need to write your own code in Python and OpenCV. You cannot use any part of the code from the link above.